Geometry and Precalculus Practice Packet

Basic Trigonometry

1.

A kite is flying 120 m above the ground. The angle of elevation from a point on the ground to the kite is $52°$.

Find the length of the kite string.

2.

A right triangle has one angle measuring $38°$ and the side opposite this angle is 14 cm.

Find:

a) the hypotenuse

b) the adjacent side

Round answers to 2 decimal places.

3.

A tower casts a shadow 48 m long. At the same time, the angle of elevation of the sun is $61°$.

Find the height of the tower.

4.

Two airplanes leave an airport at the same time. One flies 320 km due north while the other flies 450 km on a bearing of $065°$.

Find the distance between the planes.

5.

From the top of a cliff 85 m high, the angle of depression to a boat is $27°$.

Find the distance of the boat from the base of the cliff.

Volume

6.

A cylindrical water tank has radius 5 m and height 12 m.
Find its volume correct to the nearest cubic meter.

7.

A cone has height 15 cm and diameter 12 cm.
Find its volume correct to 1 decimal place.

8.

A hemisphere has radius 9 cm.
Find:

a) its volume

b) its curved surface area

Leave answers in terms of π.

9.

A rectangular swimming pool measures 18 m by 10 m and has a uniform depth of 2.4 m.
Water flows into the pool at a rate of 0.8 cubic meters per minute.
How long will it take to completely fill the pool?

10.

A sphere is melted down and recast into smaller spheres of radius 3 cm.
If the original sphere had radius 12 cm, how many smaller spheres can be made?

Literal Equations

11.

Make x the subject:

$$P = \frac{3x - 7}{2}$$

12.

Make r the subject:

$$A = \pi r^2 + 2rh$$

13.

Make x the subject:

$$\frac{a}{x} + \frac{b}{x - c} = d$$

14.

Make t the subject:

$$P = \frac{A}{1 + rt}$$

Trigonometric Equations

15.

Solve for $0° \leq x < 360°$:

$$2\cos x - 1 = 0$$

16.

Solve for $0° \leq x < 360°$:

$$2\sin^2 x - \sin x - 1 = 0$$

17.

Solve for $0° \leq x < 360°$:

$$\tan^2 x - 3\tan x + 2 = 0$$

Periodic Functions

18.

For the function

$$y = 4\sin(3x) - 2$$

find:

a) amplitude

b) period

c) vertical shift

d) maximum and minimum values

19.

Sketch two complete cycles of:

$$y = -2\cos(2x) + 1$$

Clearly label:

- amplitude

- period
- maximum points
- minimum points
- y-intercept

Logarithmic and Exponential Functions

20.

Solve the following:

$$5^{x-2} = 125$$

$$2^{x+1} = 64$$

$$\log_3(x+7) = 2$$

$$\log(x) = 2$$